CHAPTER 8 STATICS

Key idea: a body is in equilibrium when the resultant force and the resultant torque are both zero.

1 CONDITIONS OF EQUILIBRIUM

$$\sum F_x = 0 \qquad \sum F_y = 0$$

$$\sum \tau = 0 \text{ about any point}$$

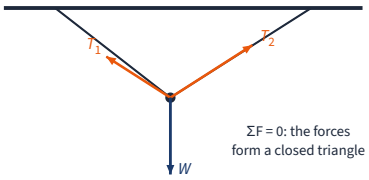
- First: translational (no linear acceleration)
- Second: rotational (no angular acceleration)

2 MOMENT OF A FORCE

$$\text{moment} = F \times d_\perp$$

- $d_\perp$: perpendicular distance from the pivot to the **line of action**
- Unit N m
- **Couple:** two equal, opposite, parallel forces; torque $= F\times$ separation; no resultant force

3 THREE FORCES



- Lines of action **concurrent** (meet at a point)
- Form a closed triangle, head to tail

$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

Lami's theorem: angle opposite each force.

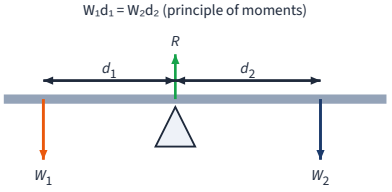
4 CENTRE OF GRAVITY

The point through which the whole **weight** appears to act.

$$\bar{x} = \frac{\sum mx}{\sum m}$$

- Uniform regular body: at its geometric centre
- Suspended freely: lies vertically below the pivot
- Same as centre of mass in a uniform field

5 PRINCIPLE OF MOMENTS



Σ clockwise moments = Σ anticlockwise moments about the same point.

Take moments about the point with the **most unknown** forces.

6 STABILITY & TOPPLING

Stable	CG rises when tilted; returns
Unstable	CG falls; moves away
Neutral	CG height unchanged

- Topples when the vertical through the CG falls **outside the base**
- Low CG and wide base give stability
- Block on a slope topples when $\tan\theta > w/h$

7 METHOD: LADDER, BEAM, HINGE

1. Draw every force: weight at the CG, reactions, friction, tension.
2. Resolve horizontally and vertically: $\sum F_x = 0, \sum F_y = 0$.
3. Take moments about a point that removes an unknown.
4. Hinge force: find both components, then $R = \sqrt{R_x^2 + R_y^2}$.

8 COMMON SECTION A TRAPS

- Using the length of the rod instead of the **perpendicular** distance.
- Forgetting the weight of the beam itself.
- Smooth wall: reaction is **perpendicular** to the wall, no friction.
- Thinking a couple can be balanced by a single force.

EXAM TIPS

- ✓Label the pivot and every distance on the diagram.
- ✓State the principle of moments in words before using it.
- ✓Check: upward forces = downward forces.
- ✓Give the direction of a hinge force as an angle.